Maximiliano Lioi

Data Scientist

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Profile

Data Scientist and Mathematical Engineer, with a Master of Science in Engineering with a specialization in Applied Mathematics. Professional experience in Bayesian modeling, data processing and analysis, statistical validation, and the development of scientific tools in Python. Strong background in machine learning, mathematical optimization, statistical inference, and applied research.

Experience

NoiseGrasp — **Feb. 2025 – present**
Data Scientist — *Santiago, Chile*

- Calibrate and validate Bayesian Marketing Mix Modeling models to estimate the incremental ROI of advertising channels and support budget allocation decisions, running ETL, training, and analysis pipelines for clients in Latin America and the U.S.
- Contribute to the development and maintenance of internal Python libraries for statistical modeling, extending analysis, validation, and reporting modules, as well as implementing improvements and fixing bugs through Git, pull requests, and code reviews.
- Implemented *Simulation-Based Calibration* (SBC) to assess the calibration of posterior distributions and build internal benchmarks of inference methods, including MCMC and variational approximations, translating the results into operational recommendations.

Universidad de O'Higgins — **Jan. 2024 – Apr. 2024**
Research Assistant — *Santiago, Chile*

- Implemented convex and nonlinear programming algorithms with *SciPy* and *CVXPY* to compute projections onto prox-regular sets, within a Fondecyt research project.
- Ran numerical experiments to verify the behavior of the implemented algorithms and support the analysis of the methods studied.

Pedro Donoso Ingenieros Asociados — **Jan. 2023 – Apr. 2023**
Data Scientist Trainee — *Santiago, Chile*

- Developed and evaluated spatial regression models to predict real estate prices, addressing spatial autocorrelation and heterogeneity with georeferenced data from the American Community Survey in R.
- Processed, cleaned, and transformed data with Python and Pandas, building explanatory variables to characterize public transportation services from ADATRAP data.

Education

Universidad de Chile — **Mar. 2024 – Jan. 2025**
Master of Science in Engineering, Specialization in Applied Mathematics, GPA: 6.9/7.0. — *Santiago, Chile*

Universidad de Chile — **Mar. 2018 – Jan. 2025**
Mathematical Engineering, GPA: 7.0/7.0. — *Santiago, Chile*

Projects

FinOpt | *Python, CVXPY, FastAPI, React/Vite, Supabase, Docker* — **Jan. 2026 – Jun. 2026**

- Designed and built a full-stack goal-based financial planning platform, integrating Monte Carlo simulation, convex optimization with probabilistic constraints (CVaR), a FastAPI backend, and a React/Vite frontend authenticated with Supabase.
- Implemented minimum investment horizon search with probabilistic guarantees, asynchronous processing of simulations/optimizations, job and result persistence, real-time progress, SQL migrations, technical documentation, and unit and integration tests. (`fin-opt web`) (`fin-opt github`)

UFC Fight Predictor | *Python, Scikit-learn, XGBoost, AutoGluon, Rich, Docker* — **Apr. 2025 – Jul. 2025**

- Built an end-to-end ML pipeline for predicting UFC fight outcomes, including ETL, feature engineering, and model training with Scikit-learn, XGBoost and AutoGluon, reaching 66% accuracy vs 55% baseline.
- Implemented a modular object-oriented architecture with reusable classes for preprocessing, evaluation, and prediction, deploying a CLI application with Rich packaged in Docker. (`ufc-predictor`) (`ufc-predictor-v2`).

Technical Skills

Languages: Python, R, Bash, SQL.
ML, Statistics, and Optimization: Pandas, NumPy, SciPy, Scikit-learn, AutoGluon, PyTorch, CVXPY, PyMC, Gurobi, Matplotlib.
Development and Frameworks: GitHub, Linux, VS Code, Claude Code, Jupyter, RStudio, Docker, FastAPI, React, Vite, Supabase.
Languages (spoken): English (B2, TOEFL ITP 2021)

Certificates

- Deep Learning with PyTorch
- Applied ML with Python
- Data Analysis with R

Publications

- Garrido, J.G., Lioi, M. & Vilches, E. *Inexact Catching-Up Algorithm for Moreau's Sweeping Processes.* Applied Mathematics & Optimization, **92**, 34 (2025). https://doi.org/10.1007/s00245-025-10307-w.